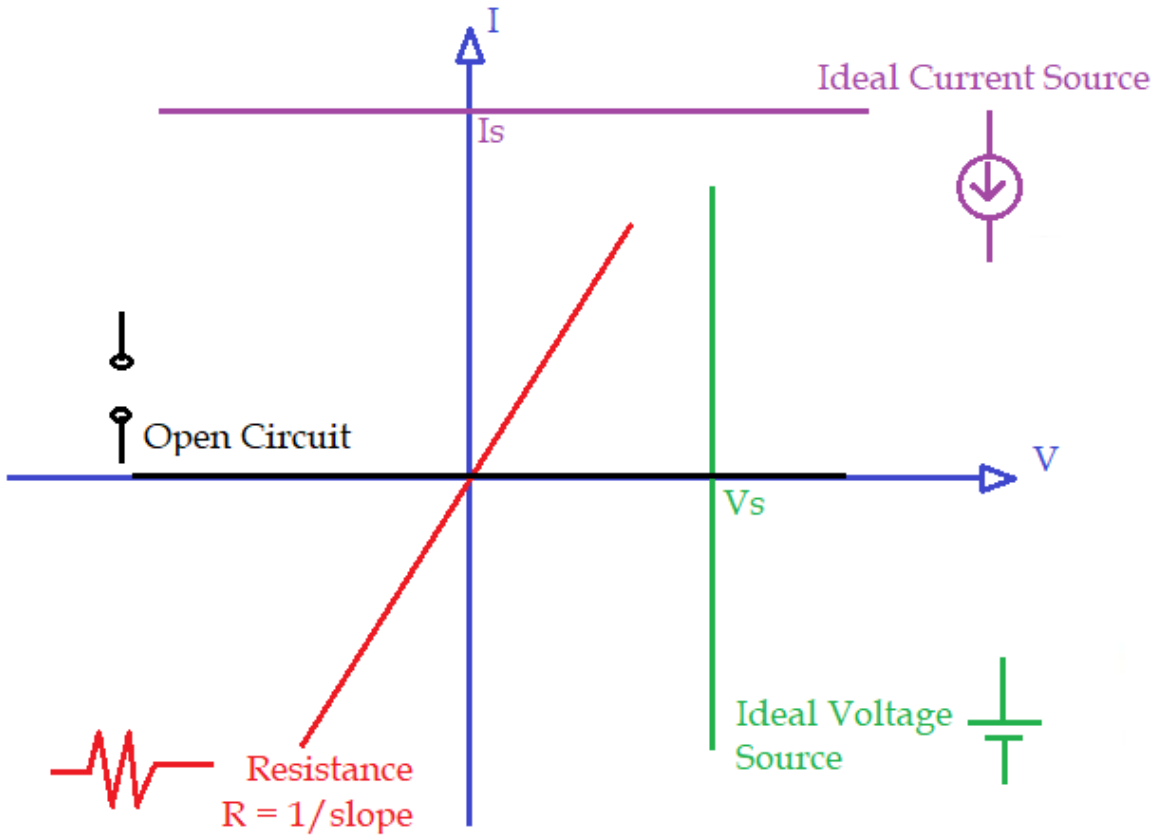
EE 312
EXP 3

LIMITING CIRCUITS

In this Experiment:

- The Voltage-Limiting Properties for Junction Diodes are Reviewed.
- The General Idea of Limiting Circuits is Introduced.
- Two Limiting (clipping) circuits are Assembled and Tested
- Brief Sample Applications are Given.

i-v Characteristics for Common Circuit Elements



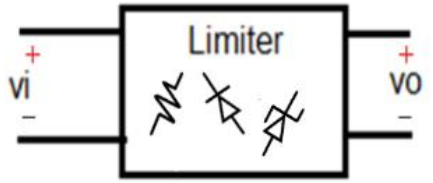
What is Voltage Limiting?

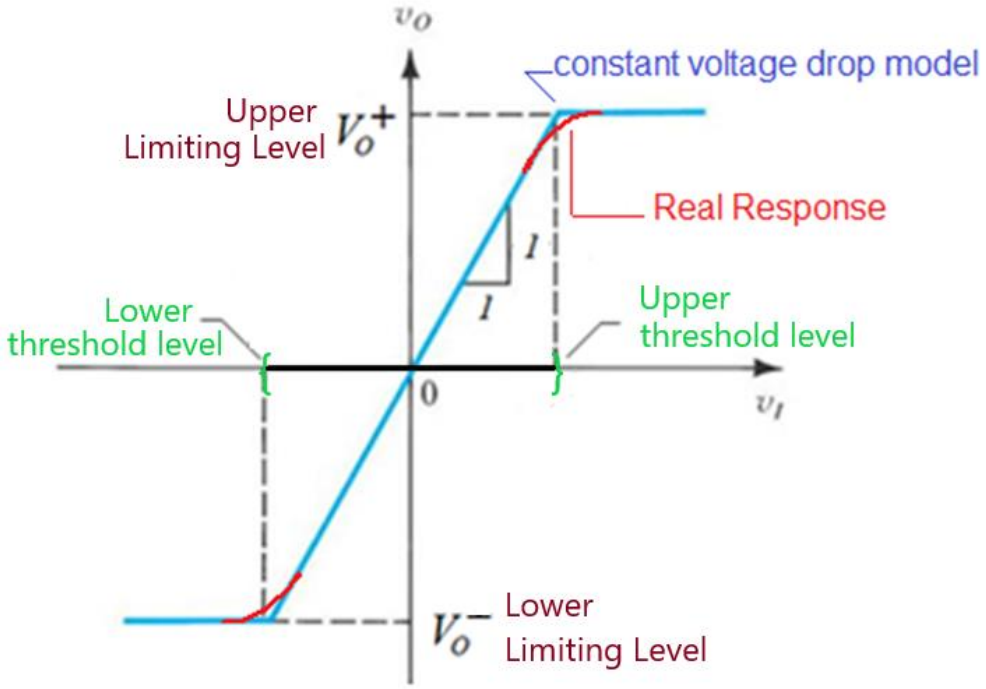
A Limiter is a circuit that has the ability to limit or clip off portions of a signal that are above or below a specified level [3].

When the input voltage magnitude is within the linear range, no current flows in the diodes, and $v_o = v_I$

When the input voltage is greater than the upper threshold level, the output voltage gets limited to V_O^+.

When the input voltage is less than the lower threshold level, the output voltage gets limited to V_O^-.

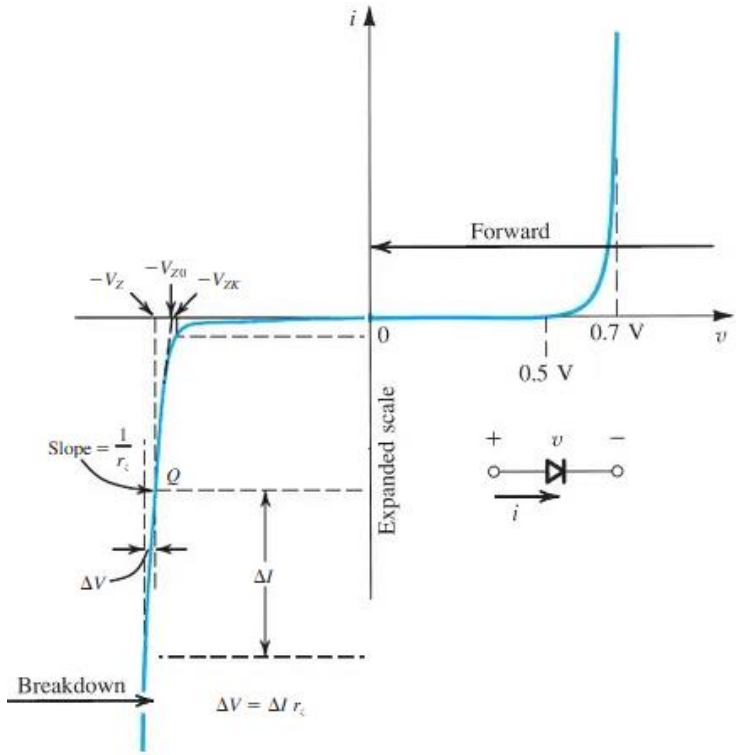




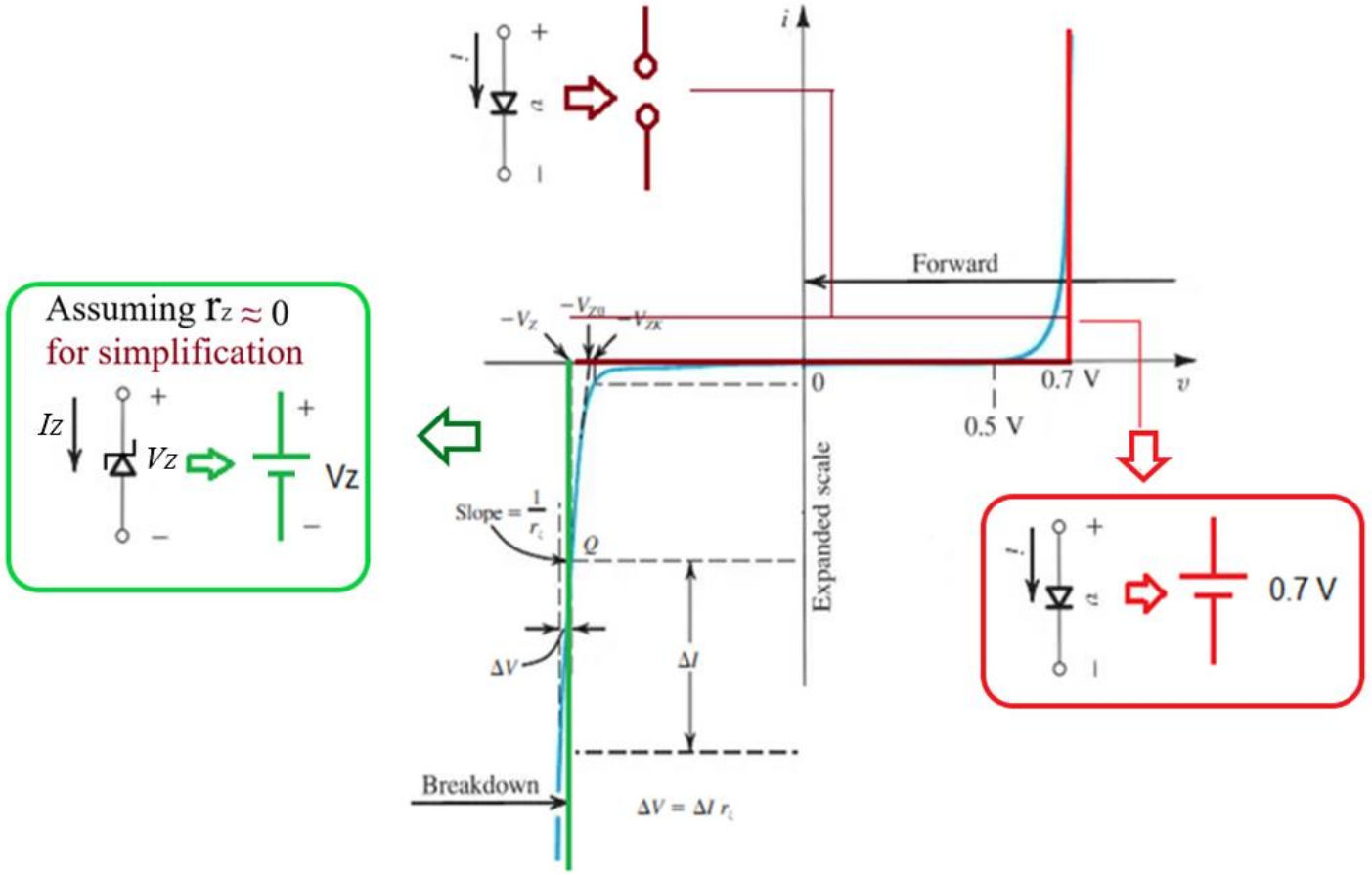
Voltage Limiting Deduced from the i-v Characteristic

A junction diode is a suitable for voltage limiting because once it conducts enough forward current its forward terminal voltage varies in a limited range: (VD ~ 0.6–0.8 V for a silicon diode).

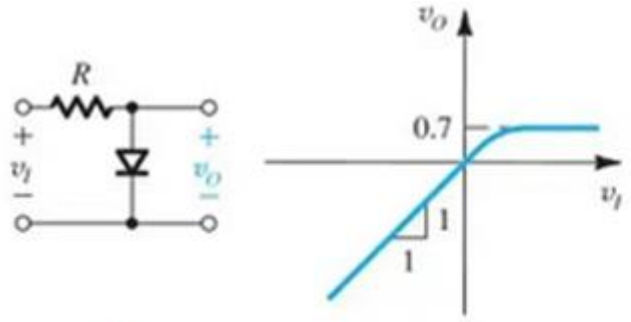
A Zener diode conducting a reverse current in the breakdown region can also be used in voltage limiting as its backward terminal voltage varies in a limited range: VZ ≅ VZ0 ± few tens of mV.



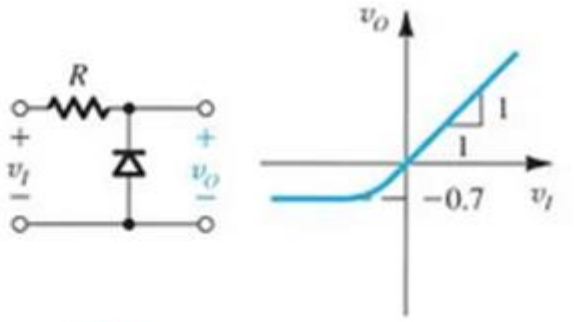
Simplified Linearized Models



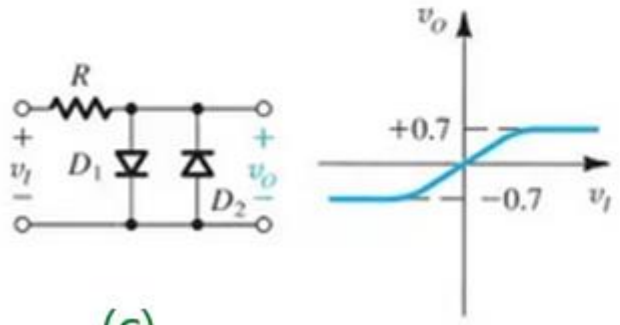
Examples of Limiting Circuits



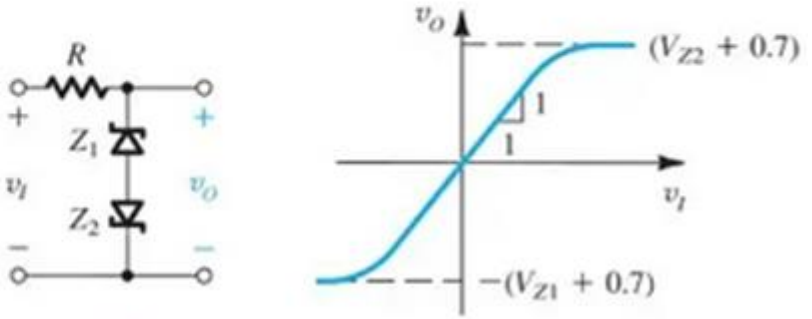
(a)



(b)

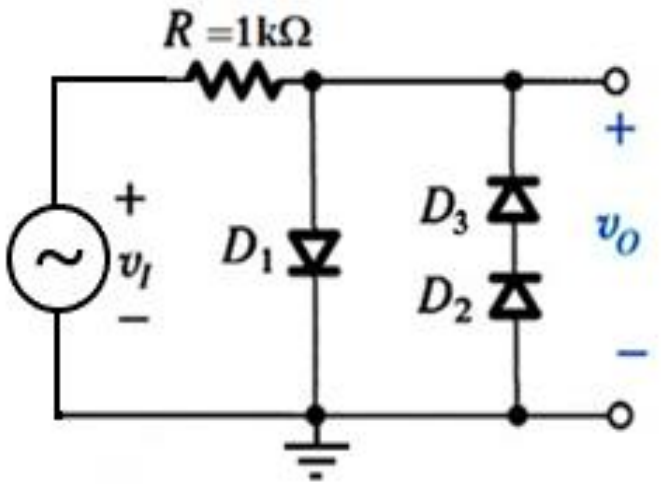


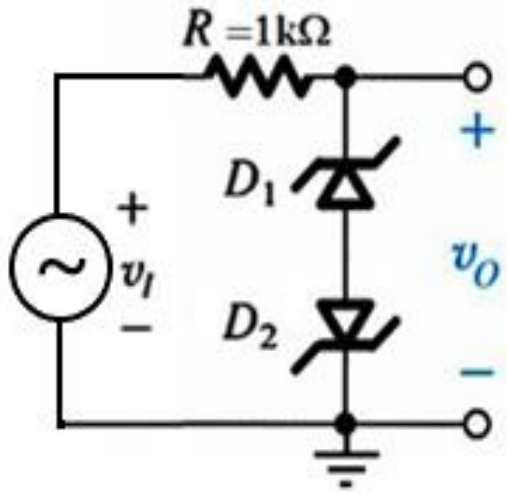
(c)

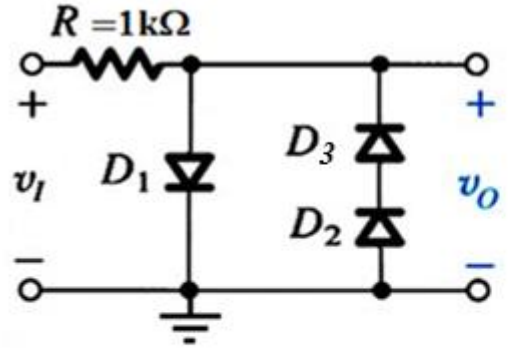


(d)

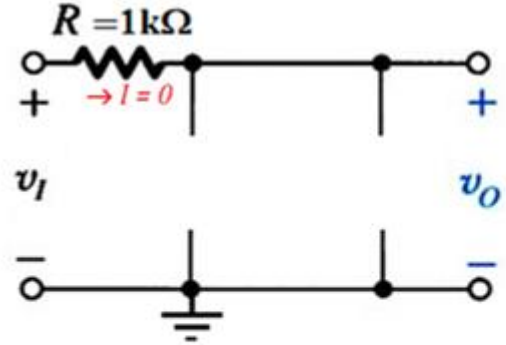
Circuits to Assembled and Tested



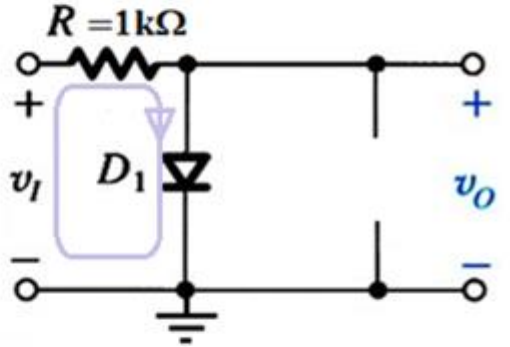


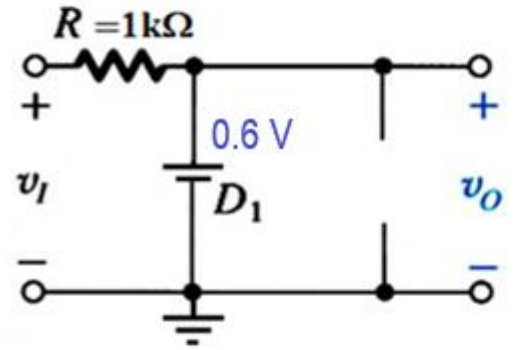


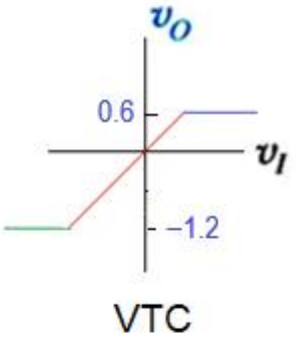
For $-1\ V < v_I < 0.5\ V$
D1 and D2 are OFF >>
$v_O = v_I$



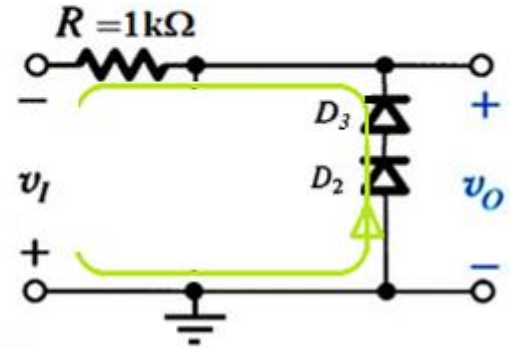
For $v_I \geq 0.5\ V$
D1 is ON and
D2 &D3 are OFF
and $v_O \approx 0.6$ V

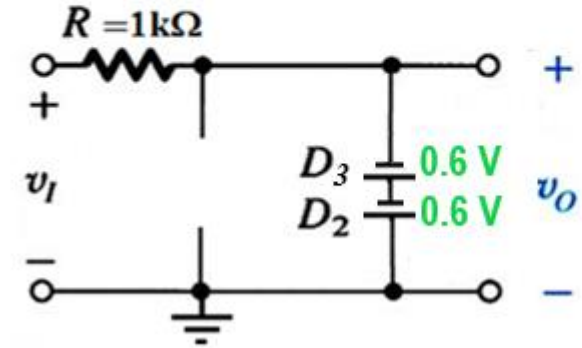


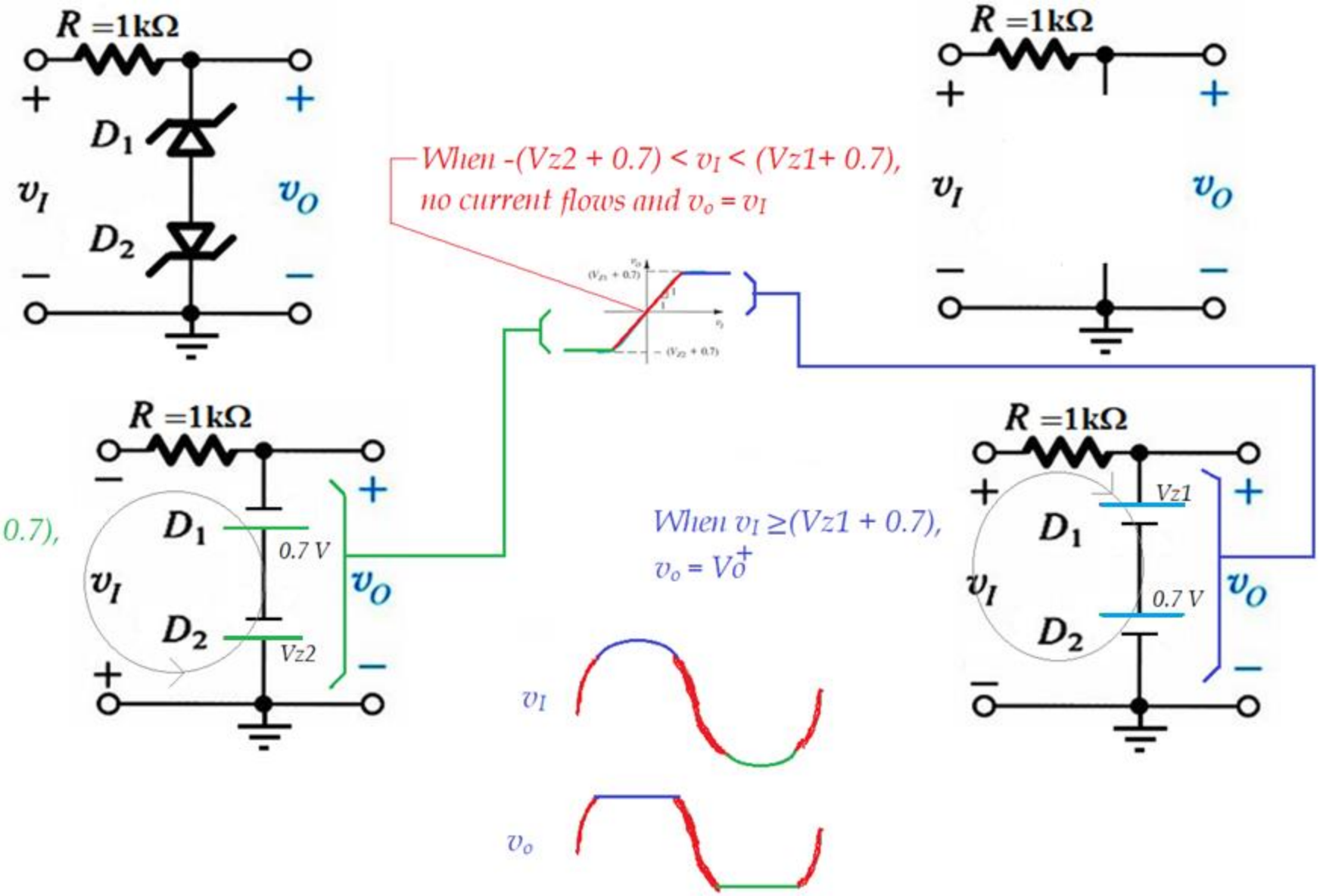




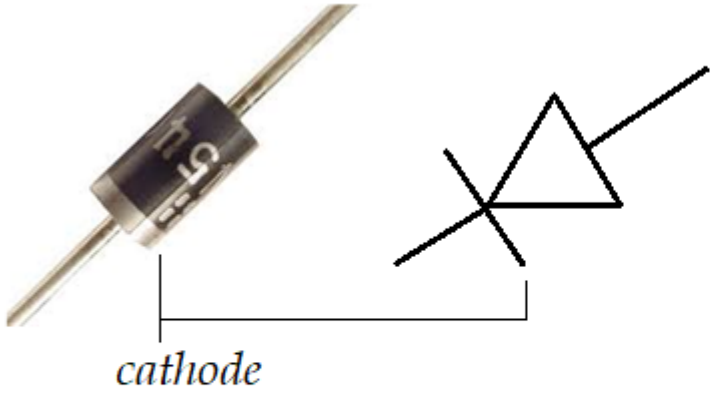
For $v_I \leq -1\ V$
D1 is OFF and
D2 & D3 are ON
and $v_O \approx -1.2\ V$



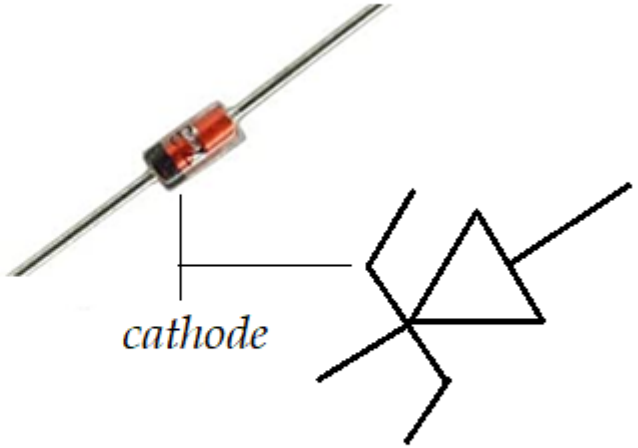



R =1kΩ
D1
D2
vI
vO
When -(Vz2 + 0.7) < vI < (Vz1+ 0.7),
no current flows and vo = vI
When vI ≤ -(Vz2 + 0.7),
vo = Vo−
0.7 V
Vz2
When vI ≥(Vz1 + 0.7),
vo = Vo+
Vz1
vI
vo

Diodes to be Used in This Experiment

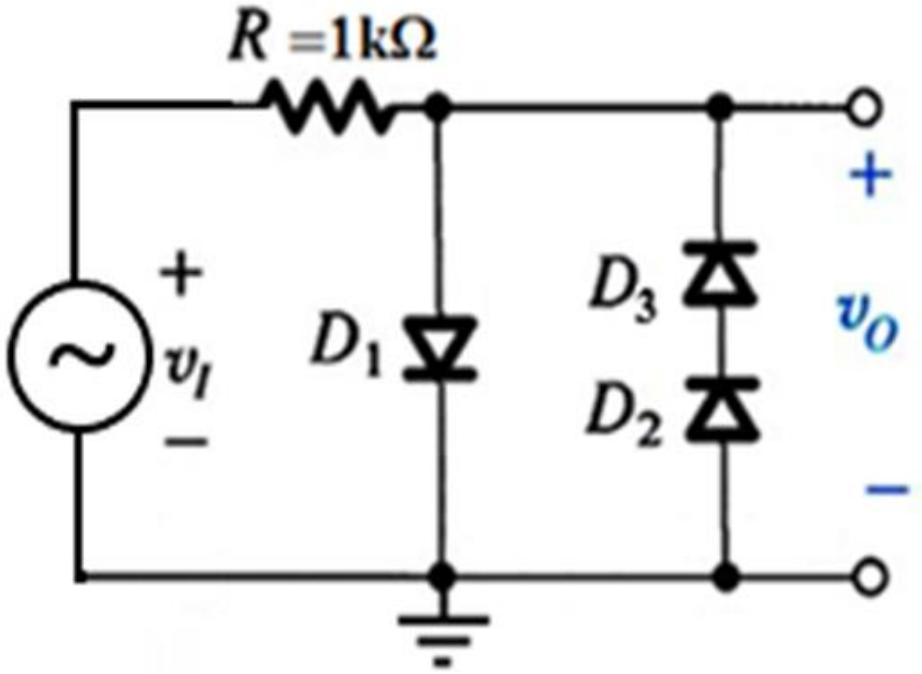


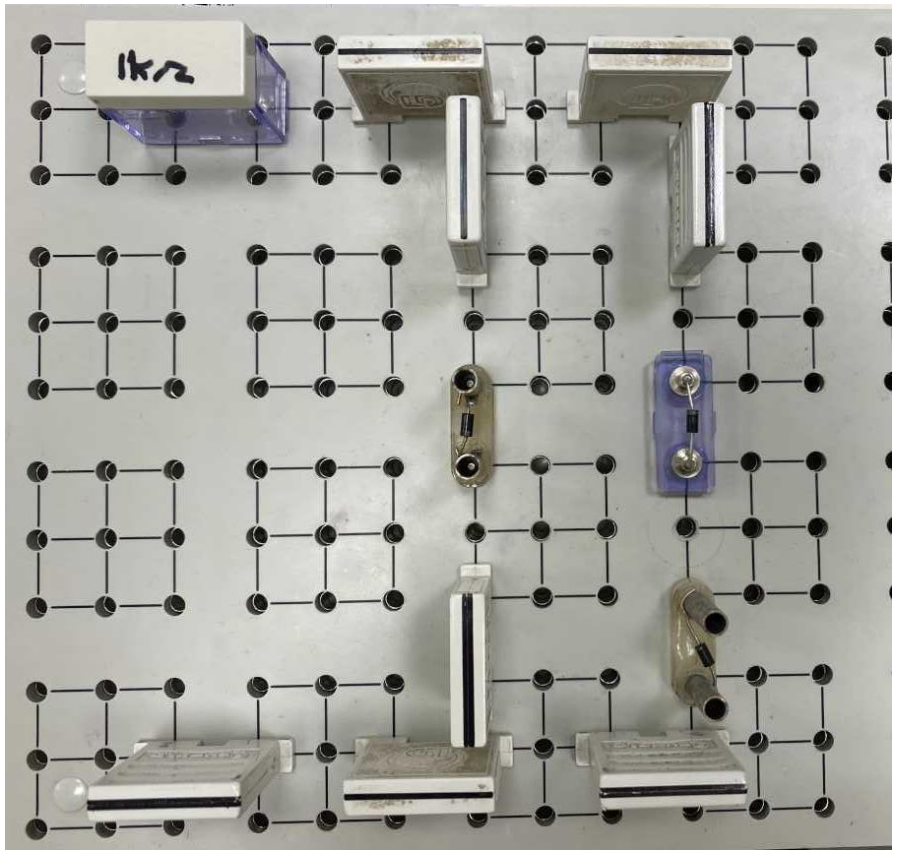
Regular Silicon Diode



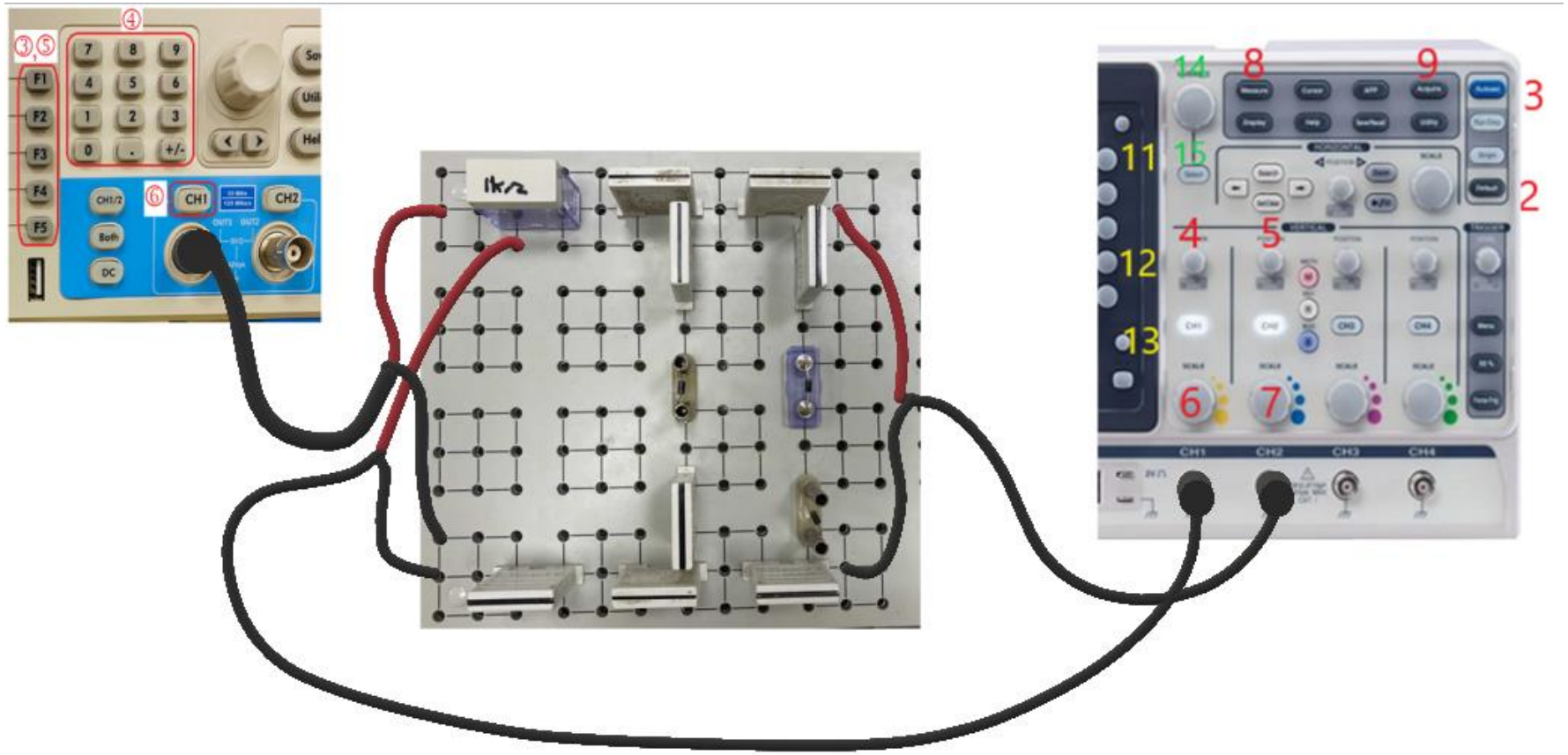
Zener Diode

Sample Implementation for Procedure A

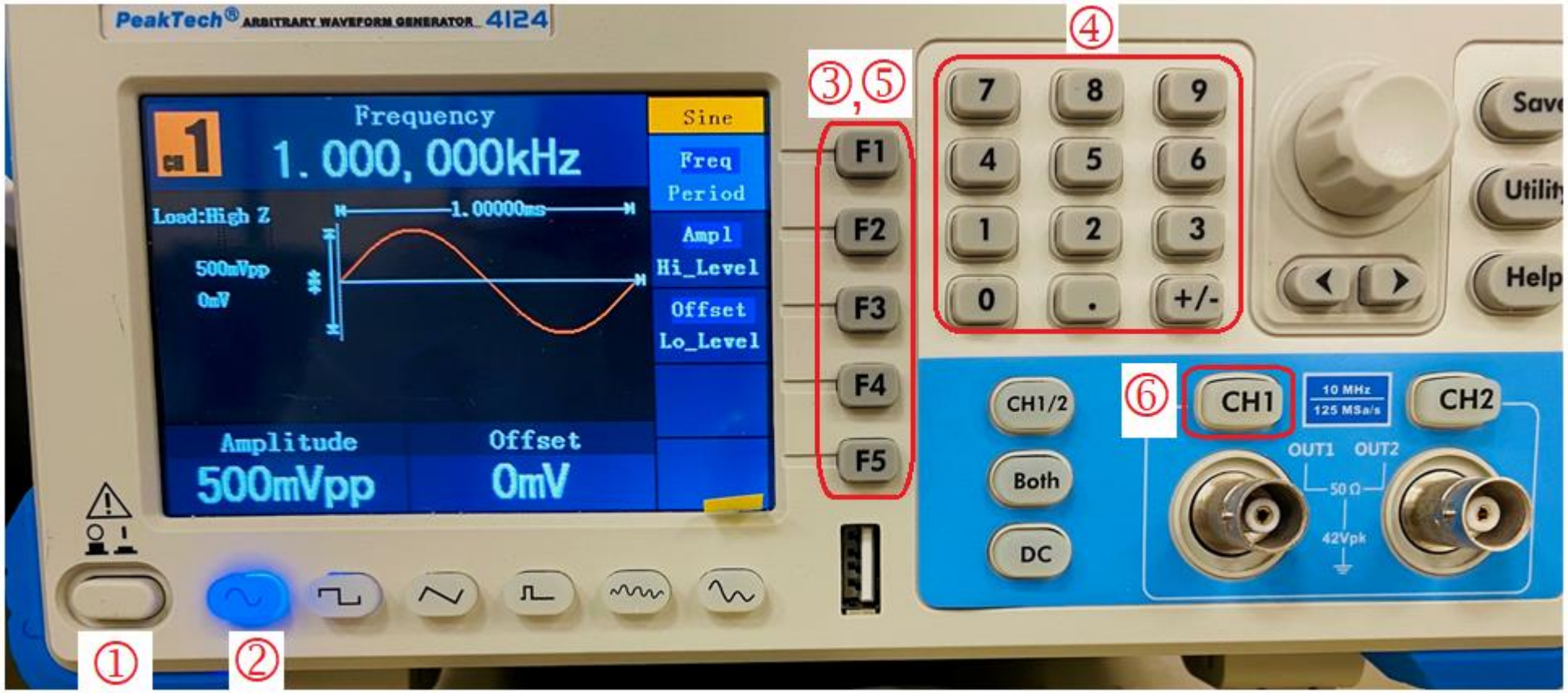




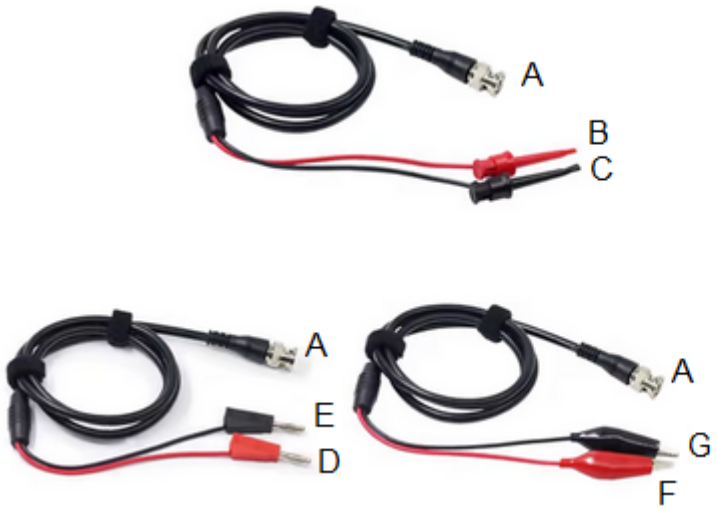
After Connecting Waveform Gen. and Scope



How to Use the Waveform Generator



Coaxial Cable Probes



BNC to Mini Grabber Hooks Cable
A. BNC Coaxial Connector
B. Mini grabber Hook
C. Ground Mini grabber Hook

BNC to Double Alligator Clips Cable
A. BNC Coaxial Connector
F. Alligator Clip
G. Ground Alligator Clip

BNC to Double Banana Plugs Cable
A. BNC Coaxial Connector
D. Banan Plug
E. Ground Banan Plug

The GwINSTEK GDS-11048 OSCILLOSCOPE

- Press the **Default** key
- Press **Autoset** key
- Press, then Adjust the Horizontal Scale Controls
- Press, then Adjust the Vertical Scale Control
- Press the **Measure** Key, then:
  - Select Add Measurement
  - Select Source1, Use the VARIABLE Key to Highlight CH1/CH2, then Press Select.
  - Select V/I, Use the VARIABLE Key to Highlight Max. Min.




LET'S DO IT!

REFERENCES

1. A. S. Sedra and K. C.Smith, *Microelectronic Circuits*, New York: Oxford University Press, 2010.
2. T. A. Almadhi, EE 312 Lab Manual
3. D. Neamen, *Microelectronics, Circuit Analysis and Design, 3rd ed.*, New York: McGraw-Hill, 2007.